The L2–L3 Bridge

A Bounded-Warrant Framework for Tracing Chip-Level Execution Conditions into AI Representations

Wiard Vasen
Independent Researcher
IJmuiden, the Netherlands

Draft v0.1 — May 2026

Abstract

Most discussions about artificial intelligence focus either on the semantic top of the computing stack — prompts, outputs, alignment — or on the physical bottom — timing channels, hardware faults, side-channel leakage. The middle of the stack, where coordinated execution turns into encoded representation, receives comparatively little systematic attention. This paper introduces the *L2–L3 bridge* as the transition where lower-layer execution conditions first become structurally capable of propagating into model behaviour. We develop a *bounded-warrant framework* — the Universal Provenance Layer (UPL) — that adjudicates claims of the form $\Gamma \vdash K \mathbin{@} B \Leftarrow W : \sigma$ under stated boundary conditions B and explicit witnesses W. We formalise the framework information-theoretically using conditional claim entropy $H(K \mid W, \Gamma, B)$, the data-processing inequality, and Fano's lower bound on adjudication error rate. We survey chip-level calibrations — mixed-precision arithmetic, TF32 mode, deterministic kernels, denormal flushing, and the thermal regime governed by Arrhenius and Johnson–Nyquist relations — whose configuration has been shown to influence model output. We postulate two complementary reverse-direction modes: reverse inference from observed AI anomalies back to candidate chip-level causes, and *toy-AI feedback*, in which a controlled small computation produces an instrumented chip-level signature, providing the framework's empirical entry point. The contribution is methodological: a bounded vocabulary for talking carefully about a zone that current AI safety and chip security literatures both leave largely untouched.

---

1 Introduction

Most discussions about AI begin at the level of outputs: answers, labels, decisions, prompts, or semantic interpretation. But AI behaviour does not emerge directly from meaning. It emerges through layers of execution.

At the bottom of the stack are physical execution conditions: cache behaviour, floating-point units, DRAM, branch predictors, memory allocation, scheduling, and timing. At the top are semantic claims: interpretations about the world. Between those two extremes lies a largely unexplored transition: the point where coordinated execution becomes encoded representation. This paper focuses on that transition. We call it the *L2–L3 bridge*.

At **L2**, execution exists as operations, kernels, and computation graphs. At **L3**, those operations crystallise into activations, weights, attention states, and encoded vectors. The central research question is:

Which lower-layer execution conditions, propagating through the L2–L3 bridge, can be witnessed, mapped, checked, and ruled out as claim-bearing for an upper-layer AI output boundary?

The claim of this work is not that chip-level events directly explain AI semantics. The claim is narrower. The L2–L3 transition *may* be the first place where lower-layer execution conditions become structurally capable of propagating into model behaviour. This transition therefore defines what we call the *bounded-warrant gap*: the zone between physical execution and semantic interpretation, where claims about AI behaviour must first become traceable, checkable, and bounded.

The contribution of this paper is methodological. We do not present new empirical results about chip-AI propagation. We introduce a vocabulary — the bounded-warrant framework — in which such results can be stated without overclaim, and a research question that delimits the relevant scope. The framework is informed in spirit by Tanenbaum's MINIX discipline [15]: small, clean, safe, understandable infrastructure beneath every claim.

Outline. Section 2 characterises the AI-on-chip stack. Section 3 introduces the bounded-warrant judgment form. Section 4 presents the UPL logic table that adjudicates status assignments. Section 5 formalises the framework information-theoretically, including Fano's lower bound on adjudication error. Section 6 concentrates on the L2–L3 bridge and presents the main figure. Section 7 surveys chip-level calibrations whose configuration is known to influence model output, including the thermal regime. Section 8 states two postulates: reverse inference from observed anomalies to chip-level candidates, and toy-AI feedback as the framework's empirical entry point. Section 9 discusses open questions.

2 The Layered AI-on-Chip Stack

We adopt a six-layer model of AI-on-chip execution. Each layer corresponds to a distinct *meaning-giving act* performed on the signal as it ascends:

- **L0 Occurrence**. FPU operations, cache lines, DRAM accesses, branch predictor state. Physical execution conditions; pre-meaning.
- **L1 Coordination**. Scheduling, memory allocation, drivers, OS. The event becomes a resource-mediated process.
- **L2 Operation**. Kernels, ops, computation graphs. The coordinated process becomes a named operation.
- **L3 Representation**. Activations, weights, attention, encoded state. The operation crystallises into encoded state.
- **L4 Decision**. Logits, tokens, labels, output commitment. The representation becomes a committed position.
- **L5 Interpretation**. Semantic claims, prompts, tasks, world meaning. The output becomes a claim about the world.

The meaning-making chain reads, from physical occurrence to semantic claim:

$$\text{Occurrence} \rightarrow \text{Coordination} \rightarrow \text{Operation} \rightarrow \text{Representation} \rightarrow \text{Decision} \rightarrow \text{Interpretation}.$$

The naming scheme is deliberately functional: each layer is named for what *happens* there, not for the technology used to implement it. PyTorch, TensorFlow, and JAX all instantiate *Operation* at **L2**, regardless of which framework is chosen. This decoupling matters for bounded warrants: a witness at **L2** need not commit to a specific framework to remain valid across re-implementations.

The reader should note where existing research traditions sit on this stack. AI safety scholarship typically operates at **L4–L5**: outputs, alignment, interpretation [3]. Chip security research typically operates at **L0–L1**: timing channels, speculative execution, microarchitectural state [6, 7]. The middle of the stack — **L2** and **L3** — is comparatively unaudited. This is the bounded-warrant gap.

3 The Bounded-Warrant Framework

The Universal Provenance Layer (UPL) introduces a judgment form that adjudicates claims about behaviour without asserting their truth.

Definition 1 (UPL judgment)**.** A UPL judgment is an expression

$$\Gamma \vdash K @ B \Leftarrow W : \sigma$$

where Γ is a set of stated assumptions, K is the claim, B is the boundary within which K is asserted, W is the witness offered in support, and $\sigma \in \Sigma$ is the status returned by adjudication.

The judgment is not a proof. UPL does not validate truth; it *adjudicates admissibility*. Adjudication returns one of nine statuses (Section 4). The framework distinguishes a candidate *UPLJudgment* (the input) from the adjudicated *UPLDecision* (the output).

For claims about how a lower-layer condition affects an upper-layer output boundary, we require a structured witness composed of six components:

Definition 2 (Layer-effect claim)**.** A layer-effect claim is a UPL judgment

$$\Gamma \vdash K_{\text{layer-effect}} @ B \Leftarrow \left(W_{\text{input}} \oplus W_{\text{exec}} \oplus W_{\text{map}} \oplus W_{\text{contract}} \oplus W_{\text{check}} \oplus W_{\text{ruleout}}\right) : \sigma$$

where the witness components are:

- W_{input} — the input or workload condition under which the claim is asserted;
- W_{exec} — observation of the lower-layer execution condition itself;
- W_{map} — the mapping from execution condition to an upper-layer model variable;
- W_{contract} — the semantic contract that fixes what counts as the upper-layer output boundary;
- W_{check} — the controlled comparison demonstrating an output-boundary shift;
- W_{ruleout} — the evidence excluding alternative attributions of the shift.

All six components are mandatory. A layer-effect claim with fewer than six receives status `incomplete`; UPL does not award partial credit. This is the discipline that prevents an execution trace alone — which contains no model information — from being claimed as supporting a behavioural conclusion.

Remark 1. Observation is not claim. Trace is not truth. No claim travels farther than its evidence.

Table 1. UPL adjudication statuses ($\sigma \in \Sigma$).

Status	Witness condition	Adjudication result
`unsupported`	No witness offered or no witness within B.	Claim is not admissible.
`observed`	W_{exec} present; other components absent.	Descriptive claim only; no behavioural conclusion.
`witnessed`	Witness components present and self-consistent.	Admissible at the witness level; no check yet.
`pending_check`	W_{check} specified but not yet executed.	Awaiting controlled comparison.
`checked`	All six components present and check succeeded.	Claim admitted within B.
`failed`	All six components present; check did not support K.	K not supported by W in B. Not equivalent to $\neg K$.
`disputed`	Conflicting witnesses or contested Γ.	Adjudication suspended pending resolution.
`incomplete`	Required witness components missing.	Cannot yet be adjudicated.
`out_of_boundary`	Witness lies outside B.	Claim falls outside stated scope.

4 The UPL Logic Table

UPL adjudicates each judgment into one of nine statuses. Table 1 states the vocabulary, its witness condition, and what may and may not follow.

The crucial distinction is between `failed` and $\neg K$. A failed check means the witness/check relation did not support K inside B; it does *not* mean K has been disproved. Negation is itself a claim and requires its own witness chain.

5 An Information-Theoretic Formulation

The qualitative status vocabulary of Section 4 admits an information-theoretic interpretation along lines familiar from Shannon [14] and the subsequent literature on mutual information [1].

5.1 *Claim entropy and witness information gain*

Let K be a binary random variable whose value is the truth of the claim under Γ and B. Before any witness is offered, the uncertainty about K is the prior claim entropy

$$H(K \mid \Gamma, B) \;=\; -\sum_{k\in\{0,1\}} \Pr(K = k \mid \Gamma, B)\, \log_2 \Pr(K = k \mid \Gamma, B).$$

A witness W reduces this uncertainty to the conditional entropy $H(K \mid W, \Gamma, B)$. The information gained by adjudicating W is the mutual information

$$I(K; W \mid \Gamma, B) \;=\; H(K \mid \Gamma, B) - H(K \mid W, \Gamma, B). \tag{1}$$

We say a witness is *claim-bearing* for K in B when $I(K; W \mid \Gamma, B) > \tau$ for some pre-stated threshold $\tau > 0$. The threshold is part of the boundary B, not a free parameter: B must specify what counts as adequate information transfer before any check is run.

Three UPL statuses correspond naturally:

- `checked`: $I(K; W) > \tau$ and the sign of the update supports K.
- `failed`: $I(K; W) > \tau$ and the sign of the update opposes K.
- `incomplete`: $I(K; W) < \tau$.

The remaining statuses correspond to violations of well-formedness (boundary, consistency, completeness) and are not reducible to information-theoretic quantities alone.

5.2 *Inter-layer mutual information*

Let X_i denote the random variable describing the state at layer **Li**. For a propagation chain $X_0 \to X_1 \to \cdots \to X_5$, the data-processing inequality [1] gives

$$I(X_0; X_5) \;\le\; \min_{0 \le i < 5} I(X_i; X_{i+1}). \tag{2}$$

The bound says: the amount of **L0** information reaching **L5** cannot exceed the narrowest channel along the path. If any single inter-layer channel has low mutual information, no fine-grained **L0** condition can survive to **L5**.

5.3 *The L2–L3 transition as channel*

Within this framing, the L2–L3 transition is itself a channel, denoted $\mathcal{C}_{2\to 3} : X_2 \to X_3$, whose capacity $C(\mathcal{C}_{2\to 3})$ bounds how much **L2**-level operational detail can be encoded into **L3**-level representational state.

Proposition 1 (Bridge bottleneck)**.** For a propagation chain $X_0 \to \cdots \to X_4$ in which $\mathcal{C}_{2\to 3}$ is the unique non-trivial bottleneck,

$$I(X_0; X_4) \;\le\; C(\mathcal{C}_{2\to 3}).$$

Proposition 1 is a corollary of (2) and provides a precise sense in which the L2–L3 bridge is informationally decisive. Where the bridge is narrow — e.g. aggressive quantisation, dimensionality collapse, attention masking — chip-level conditions cannot propagate finely. Where it is wide — e.g. high-precision matmul with preserved fan-out — they can. The framework does not assert either condition holds in any specific deployment; it states the structural relation.

5.4 *Lower bound on error rate from Fano's inequality*

Information-theoretic constraints also give us a lower bound on *how often* an adjudication can be in error. Fano's inequality [1, 2] relates the conditional entropy $H(K \mid W)$ to the probability P_e of incorrect adjudication. For a binary claim $K \in \{0, 1\}$:

$$H(K \mid W, \Gamma, B) \;\le\; H_b(P_e), \tag{3}$$

where $H_b(p) = -p \log_2 p - (1-p)\log_2(1-p)$ is the binary entropy function. Equivalently,

$$P_e \;\ge\; H_b^{-1}\big(H(K \mid W, \Gamma, B)\big), \tag{4}$$

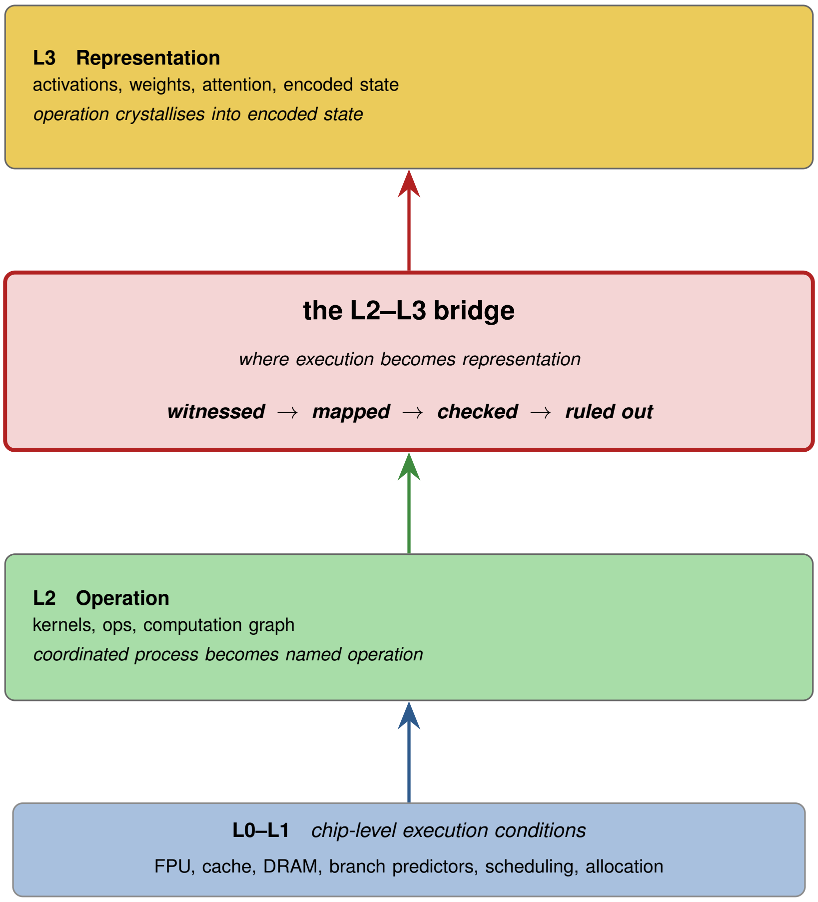

Figure 1. The L2–L3 bridge as a bounded-warrant zone. Lower-layer chip-level execution conditions (**L0–L1**) traverse **L2** *Operation* before crystallising at **L3** *Representation*. The bridge is the locus at which a chip-level event becomes structurally capable of carrying a claim. The four operations *witnessed* → *mapped* → *checked* → *ruled out* summarise the bounded-warrant discipline that any layer-effect claim must satisfy across this zone.

where H_b^{-1} denotes the inverse of H_b on $[0, 1/2]$.

The interpretation is operational: if the witness leaves the claim sufficiently uncertain — that is, if $H(K \mid W, \Gamma, B)$ is large — then no adjudication procedure can drive P_e below the bound. Adjudicating a claim with a weak witness will be wrong some of the time, and the lower bound on that error rate is set by information, not by adjudicator ingenuity. This is one of the strongest formal motivations for the bounded-warrant discipline: constructing W with high mutual information is not optional, because P_e is rate-limited by $H(K \mid W)$ regardless of how the adjudication is implemented.

Remark 2. Fano's bound applies to the adjudicator viewed as a decoder of K from W. UPL is not a decoder in the cryptographic sense, but the structural relation holds: insufficient information cannot be compensated for by procedure.

6 The L2–L3 Bridge

Figure 1 situates the bridge within the stack and labels the bounded-warrant framework as a four-stage discipline.

Three properties make the L2–L3 transition the natural locus for bounded warrants:

1. **First point of representational identity.** Below **L3**, no entity in the computation is identifiable as "a weight" or "an activation". There is only the framework's view of operations on tensors. At **L3**, those tensors acquire positional and semantic role within the model.

2. **Bottleneck candidacy.** Per Proposition 1, the **L2-L3** channel is a candidate bottleneck for information propagation. Whether it is the binding constraint depends on the specific deployment, but it is the layer at which precision, quantisation, and architectural choices most directly intersect.

3. **Witness constructibility.** Witnesses at **L0** are abundant (hardware counters, performance traces, microarchitectural probes) but often disconnected from model semantics. Witnesses at **L5** are abundant (output logs, evaluations, prompts) but disconnected from physical execution. The L2–L3 bridge is the layer at which a witness can plausibly connect both ends — where W_{exec} and W_{map} can be defined together.

Figure 2 extends the bridge model with explicit operator structure and bidirectional reversibility. The L2.5 zone is not a new physical layer; it is a bounded reasoning zone within which forward and reverse claims can be named, measured, and adjudicated under the same witness discipline.

7 Chip-Level Calibrations and Their Propagation

The bounded-warrant framework is motivated by, but does not assume, real instances of chip-level conditions affecting model behaviour. Several configurable *calibrations* are documented in the literature as inducing observable changes at the model output. We list them as candidate domains for which the framework should ultimately be tested.

Mixed-precision arithmetic (FP32 $\leftrightarrow$ FP16/BF16). Mixed-precision training and inference reduce numerical precision of intermediate values, typically with master-weight preservation [8]. The chip-level effect is a change in floating-point rounding behaviour; the L2 effect is a change in kernel selection; the L3 effect is a perturbation of activations whose magnitude depends on operator dynamic range. The witness chain $W_{\text{input}} \oplus W_{\text{exec}} \oplus W_{\text{map}} \oplus W_{\text{contract}} \oplus W_{\text{check}} \oplus W_{\text{ruleout}}$ is in principle constructible for any specific input on which the two precisions diverge in output.

Tensor Float-32 (TF32) on NVIDIA Ampere. TF32 [10] retains FP32 dynamic range but truncates the mantissa to 10 bits, accelerating matmul at the cost of a small precision reduction in the inner-product. TF32 is enabled by default in PyTorch on Ampere-class hardware unless explicitly disabled. The configuration — a single flag in the runtime — is a chip-level calibration in the sense developed here: a property of the execution layer that may or may not be relevant to a given output boundary.

Deterministic versus non-deterministic kernels. Several deep-learning kernels admit multiple implementations that return numerically distinct outputs on identical inputs due to non-associativity of floating-point addition under varying reduction orders [9]. cuDNN, for instance, exposes a deterministic-mode flag that constrains kernel choice. The chip-level effect

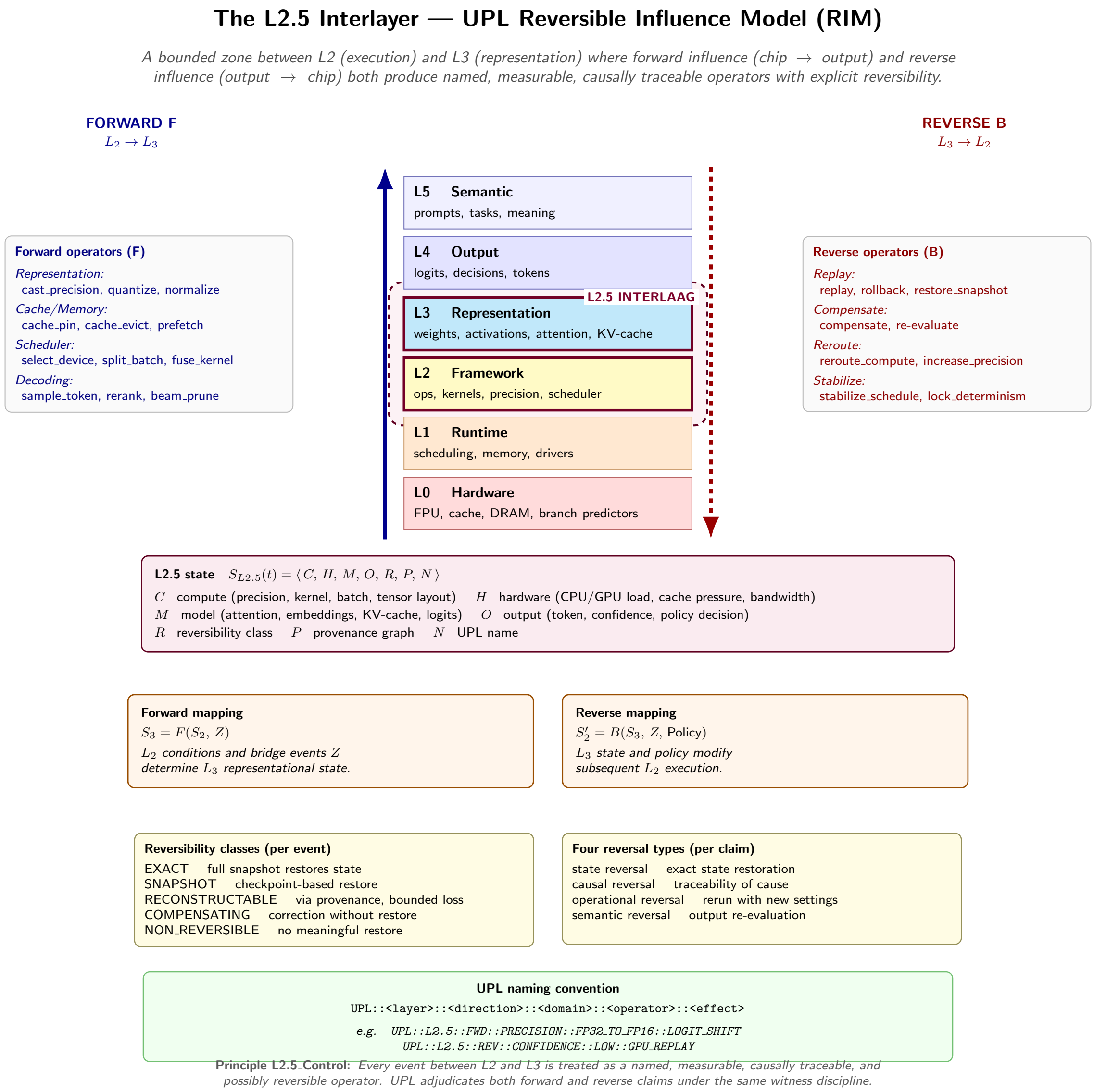

Figure 2. The **L2**.5 Interlayer — UPL Reversible Influence Model (RIM). The bounded zone between **L2** (execution) and **L3** (representation) is where forward influence (chip → output) and reverse influence (output → chip) both produce named, measurable, causally traceable operators with explicit reversibility. The L2.5 state $S_{L2.5}(t) = \langle C, H, M, O, R, P, N \rangle$ captures compute, hardware, model, output, reversibility class, provenance, and UPL name. Forward and reverse mappings share the same six-witness discipline. Every event between **L2** and **L3** is treated as a named, measurable, causally traceable, and possibly reversible operator (*Principle L2.5_Control*).

is a change in reduction order; the L3 effect is a perturbation in computed activations; the L4 effect in rare cases is a flipped classification near decision boundaries.

Denormal flush-to-zero. Many FPUs offer a *flush-to-zero* mode in which subnormal floating-point values are replaced by zero. The chip-level effect is a change in rounding behaviour for very small values. Propagation through **L2–L3** depends on whether training or inference produces denormals in the relevant operator, which itself depends on architecture and initialisation.

Memory layout and kernel autotuning. Frameworks such as cuDNN run an autotuning phase that selects kernel implementations based on input shape and device state. The choice may differ across runs, leading to identical inputs producing slightly different outputs. The relevant calibration is at **L1–L2**: the autotuner's selection policy.

Thermal regime and operating temperature. A chip's operating temperature T is a continuous calibration with well-documented influence on lower-layer error rates. Two physical mechanisms apply. First, Johnson–Nyquist noise gives a thermal floor on the voltage fluctuation across any resistive element of resistance R in bandwidth B:

$$\langle v_n^2 \rangle \;=\; 4k_B T R B, \tag{5}$$

where k_B is the Boltzmann constant. Higher T raises the noise floor, reducing margin against signal-discrimination errors. Second, thermally activated failure rates — charge-leakage, single-event upsets, and certain refresh-related DRAM errors — follow an Arrhenius dependence:

$$\lambda(T) \;=\; \lambda_0 \,\exp\!\left(-\frac{E_a}{k_B T}\right), \tag{6}$$

where λ_0 is a baseline rate and E_a is the activation energy of the relevant failure mode. Large-scale field studies of production DRAM [12] confirm that error rates rise strongly with operating temperature, and architectural studies have documented disturbance-error mechanisms whose rate depends on both temperature and access pattern [5]. Within the present framework, T is a chip-level calibration whose configuration is partially within operator control (cooling, workload placement, throttling policy) and partially outside it (ambient temperature, transient bursts). A layer-effect claim that implicates a thermal regime would require W_{exec} to include a temperature trace and W_{ruleout} to exclude attribution to concurrent calibrations.

Each calibration above is, in UPL terms, a hypothesis about a candidate signal. None is here claimed to be a witnessed, mapped, checked, and ruled-out cause of any specific AI behaviour under any specific boundary. The contribution of this paper is the vocabulary in which such claims could be made and adjudicated. The bounded foundational claim of UPL [4, 13] is precisely that: for specific behaviours, under stated boundaries, with explicit mapping and check, a chip-level trace *can* be made claim-bearing for an upper-layer output. The reduction is local, bounded, witnessed.

8 Postulates of Reverse Inference and Toy-AI Feedback

The framework presented so far runs in the forward direction: from a stated chip-level calibration W_{exec} through the bridge to an upper-layer output boundary W_{check}. Two symmetric questions arise. The first concerns *reverse inference* — tracing from an observed AI anomaly back to a candidate chip-level cause. The second concerns *toy-AI feedback* — the observation that a small, controlled AI computation itself produces a measurable signature at the chip level, which provides an empirical entry point to the framework.

8.1 Reverse inference

> *Given an observed anomaly at the AI output boundary, can the same witness vocabulary be used in reverse, to identify a candidate chip-level execution condition that may have produced it?*

We do not assume this is possible in general. We postulate it as a research direction, subject to the same bounded-warrant discipline as the forward direction.

Postulate 1 (Reverse inference)**.** Let K^* be a claim of the form "output anomaly A at **L4** was carried by chip-level condition $c \in \mathcal{C}$" where $\mathcal{C}$ is a finite catalogue of candidate calibrations. Then under appropriate assumptions Γ, a reverse-direction UPL judgment

$$\Gamma \ \vdash \ K^* \ @ \ B \ \Leftarrow \ \left(W_{\text{anomaly}} \oplus W^*_{\text{exec}} \oplus W^*_{\text{map}} \oplus W_{\text{contract}} \oplus W^*_{\text{check}} \oplus W^*_{\text{ruleout}}\right) \ : \ \sigma$$

may be adjudicated with the same status vocabulary Σ as the forward direction, provided:

- the anomaly is observable at **L4** (or higher) and admits a stated boundary B;
- the candidate catalogue $\mathcal{C}$ is finite, enumerated, and disjoint;
- the inverse map W^*_{map} can be constructed from **L3** representational state back to a specific candidate $c \in \mathcal{C}$;
- W^*_{ruleout} excludes the remaining candidates in $\mathcal{C} \setminus \{c\}$ under B.

A useful concrete form for W^*_{map} is Bayesian. If $\mathcal{C} = \{c_1, \ldots, c_n\}$ with prior $\Pr(c_i)$, the posterior over candidates given the observed anomaly A is

$$\Pr(c_i \mid A, \Gamma, B) \ = \ \frac{\Pr(A \mid c_i, \Gamma, B) \ \Pr(c_i)}{\sum_{j=1}^{n} \Pr(A \mid c_j, \Gamma, B) \ \Pr(c_j)}. \tag{7}$$

A reverse-direction judgment reaches `checked` for candidate c_i when the posterior concentrates on c_i above a stated threshold, and when W^*_{ruleout} supplies witnesses that the remaining candidates are excluded under B. Equation (7) is not a procedure but a constraint on what counts as a defensible reverse-inference judgment.

The postulate is consequential. It says the bounded-warrant framework is in principle bi-directional, even though the information-theoretic challenges are asymmetric. Forward inference faces the data-processing inequality (2): information may be lost in propagation, so a strong chip-level signal may not survive to be witnessed at **L4**. Reverse inference faces non-identifiability: many chip-level events may produce the same observable anomaly, so W^*_{ruleout} must work harder.

Reverse inference connects naturally to causal-inference frameworks [11] and to the literature on information-flow tracking in security [6]. We do not extend the formal connection in this paper. We note only that the bounded-warrant vocabulary is symmetric in form, and asymmetric only in the difficulty of constructing the required witnesses.

8.2 Toy-AI feedback as an empirical entry point

Reverse inference (Postulate 1) carries a constructive obstacle: how is W^*_{map} to be defined in practice? The forward direction draws on hardware counters and framework instrumentation; the reverse direction needs a procedure that takes an observed output anomaly and produces a posterior over chip-level candidates. We postulate that the simplest tractable setting in which this map is constructible is the case of a deterministic toy AI.

Postulate 2 (Toy-AI feedback)**.** Let $\mathcal{M}$ be a deterministic, fully specified computation — a *toy AI* — confined within hardware boundary B. Let $X_3(\mathcal{M}, t)$ denote $\mathcal{M}$'s **L3** representational state at time t, and let $S(\mathcal{M}, t) \in \mathcal{S}$ denote the chip-level signature produced as a by-product of $\mathcal{M}$'s execution — comprising thermal trajectory, instantaneous power draw, performance-counter readings, cache footprint, and memory-bus timing. Then under appropriate Γ and over a stated horizon,

$$I\big(S(\mathcal{M},t)\,;\,X_3(\mathcal{M},t)\,\big|\,\Gamma,B\big) \;>\; 0. \tag{8}$$

The postulate asserts that the chip-level signature is non-trivially informative about the toy AI's internal state. This is not a strong claim — the right-hand side need not be large — but it is sufficient to make reverse inference non-vacuous in at least one setting. By Fano's inequality (3) applied in the reverse direction, a non-zero mutual information lower-bounds the achievable adjudication accuracy from below; conversely, an instrumented toy-AI setting provides a concrete construction of W^*_{exec}, W^*_{map}, and ultimately W^*_{ruleout}.

The toy-AI setting is empirically tractable in a way that production-scale systems are not. The model is small enough to be fully characterised; the hardware boundary B is small enough to be instrumented; the computation is deterministic enough that repeated execution yields a statistical estimate of $I(S;X_3)$. The framework's reverse-inference postulate is therefore reduced, in its first instantiation, to a controlled measurement problem: can a chip-level signature be shown to carry τ bits of information about a toy AI's representational state within stated B?

Remark 3. Postulate 2 does not assume the converse for production systems. Scaling from toy-AI feedback to bounded reverse inference on production AI is left open. The postulate serves as the framework's empirical entry point, not as a universal claim.

9 Discussion

Scope and first witnessed instance. This paper presents the framework within which layer-effect claims can be made. The next research step described in earlier drafts — the construction of a single, fully witnessed instance of a layer-effect claim across the **L2–L3** bridge — has now been executed.

Under a local-machine, single-venv boundary using PyTorch 2.12.0 and a deterministic toy binary linear classifier ($d = 16$), same-precision determinism held in both fp32 and fp16. The cross-precision contrast yielded zero control flips and one margin flip. Under the declared boundary B, the UPL layer-effect claim

> *"Changing only the execution precision from fp32 to fp16 in the **L2** dot-product operation changes the measured margin representation $z = \mathbf{w}\cdot\mathbf{x} + b$ enough to cross the predefined binary decision boundary for at least one input in X_{margin}, while X_{control} remains invariant"*

was adjudicated with $\sigma = \texttt{checked}$. The full witness chain $\big(W_{\mathrm{input}} \oplus W_{\mathrm{exec}} \oplus W_{\mathrm{map}} \oplus W_{\mathrm{contract}} \oplus W_{\mathrm{check}} \oplus W_{\mathrm{ruleout}}\big)$ is available in the project repository [16].

This establishes the feasibility of the UPL witness discipline in a toy precision-contrast setting. It does not establish broader claims about large language models, production AI systems, or hardware behaviour generally. The bounding is intentional and follows directly from the framework's discipline.

Relation to existing fields. UPL is not an AI safety framework. It is not a chip security framework. It is a provenance vocabulary that may inform both. AI safety has tools for talking about output behaviour; chip security has tools for talking about microarchitectural state.

Neither field, at present, has a vocabulary for adjudicating claims that link the two. This paper proposes one. Whether the vocabulary is adopted depends on whether the bounded warrants it demands can in fact be constructed.

What this paper does not claim. The framework does not assume that chip-level conditions matter for AI semantics. It does not assert that any specific calibration in Section 7 causes any specific behaviour under any specific boundary. It does not claim that reverse inference is achievable in any specific case. It claims only that a bounded vocabulary exists in which such claims could be stated, adjudicated, and limited.

Open questions. At least four:

- What measurement instruments can construct W_{map} across the L2–L3 bridge for current production frameworks?
- Under what conditions is W_{ruleout} constructible without exhaustive enumeration of $\mathcal{C}$?
- How does the framework compose across multiple bridges (e.g. **L0-L2** and **L2-L3** jointly)?
- Does the reverse-inference postulate (Postulate 1) admit a constructive proof under bounded assumptions, or does it require case-by-case argument?

A note on terminology. We have deliberately avoided the language of *fault*, *attack*, and *breach* throughout. These terms carry witness commitments that the present framework does not yet support. A fault asserts a failure of specification; an attack asserts intent and control; a breach asserts a witnessed crossing of a security boundary. The framework presented here talks instead of *execution conditions*, *candidate signals*, and *layer-effect claims*. The terminological restraint is part of the bounded-warrant discipline.

Acknowledgements

This work proceeds in the spirit of small, clean, safe, understandable infrastructure that has marked Andrew Tanenbaum's contributions to operating systems pedagogy. Errors and infelicities remain the author's own.

References

[1] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, 2nd edition, 2006.

[2] Robert M. Fano. Transmission of information: A statistical theory of communications. *MIT Press*, 1961.

[3] Dan Hendrycks, Nicholas Carlini, John Schulman, and Jacob Steinhardt. Unsolved problems in ML safety, 2021.

[4] Sara Hooker. The hardware lottery, 2020.

[5] Yoongu Kim, Ross Daly, Jeremie Kim, Chris Fallin, Ji Hye Lee, Donghyuk Lee, Chris Wilkerson, Konrad Lai, and Onur Mutlu. Flipping bits in memory without accessing

them: An experimental study of DRAM disturbance errors. In *International Symposium on Computer Architecture (ISCA)*, 2014.

[6] Paul Kocher, Jann Horn, Anders Fogh, Daniel Genkin, Daniel Gruss, Werner Haas, Mike Hamburg, Moritz Lipp, Stefan Mangard, Thomas Prescher, Michael Schwarz, and Yuval Yarom. Spectre attacks: Exploiting speculative execution. In *IEEE Symposium on Security and Privacy*, 2019.

[7] Moritz Lipp, Michael Schwarz, Daniel Gruss, Thomas Prescher, Werner Haas, Anders Fogh, Jann Horn, Stefan Mangard, Paul Kocher, Daniel Genkin, Yuval Yarom, and Mike Hamburg. Meltdown: Reading kernel memory from user space. In *USENIX Security Symposium*, 2018.

[8] Paulius Micikevicius, Sharan Narang, Jonah Alben, Gregory Diamos, Erich Elsen, David Garcia, Boris Ginsburg, Michael Houston, Oleksii Kuchaiev, Ganesh Venkatesh, and Hao Wu. Mixed precision training. In *International Conference on Learning Representations*, 2018.

[9] Prabhat Nagarajan, Garrett Warnell, and Peter Stone. Deterministic implementations for reproducibility in deep reinforcement learning. In *AAAI Workshop on Reproducible AI*, 2019.

[10] NVIDIA Corporation. NVIDIA A100 Tensor Core GPU Architecture. Technical White Paper, 2020.

[11] Judea Pearl. *Causality: Models, Reasoning and Inference*. Cambridge University Press, 2nd edition, 2009.

[12] Bianca Schroeder, Eduardo Pinheiro, and Wolf-Dietrich Weber. DRAM errors in the wild: A large-scale field study. In *ACM SIGMETRICS Performance Evaluation Review*, 2009.

[13] D. Sculley, Gary Holt, Daniel Golovin, Eugene Davydov, Todd Phillips, Dietmar Ebner, Vinay Chaudhary, Michael Young, Jean-François Crespo, and Dan Dennison. Hidden technical debt in machine learning systems. In *Advances in Neural Information Processing Systems*, 2015.

[14] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948.

[15] Andrew S. Tanenbaum. *Operating Systems: Design and Implementation*. Prentice-Hall, 1987.

[16] Wiard Vasen. Demo_06: One bounded witnessed instance of a layer-effect claim under upl discipline. https://github.com/wiard/upl-foundation/tree/demo_06-checked-v1.0/demo_06, 2026. Bounded UPL Foundation artefact; $\sigma =$ checked under stated boundary B.